

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	15-07-2026
Team ID	SWTID-2026-1581
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Data Submission	- Enter personal details (age, education, marital status, family size) - Enter financial & employment details (income, employment type, years employed) - Enter housing & asset details (own car, own property, housing type)
FR-2	Data Preprocessing	- Validate and clean form inputs - Encode categorical fields using the trained label encoders - Derive engineered features (e.g. years employed from raw values)
FR-3	Credit Risk Prediction	- Load the trained model (model.pkl) - Run inference on the encoded feature vector - Classify the applicant as low risk (Approved) or high risk (Rejected)
FR-4	Result Communication	- Display an Approved / Rejected result page - Show a short, friendly explanation of the outcome
FR-5	Model Lifecycle Management (Admin)	- Retrain the model on updated historical data (application_record.csv, credit_record.csv) - Compare model candidates (Logistic Regression, Decision Tree, Random Forest, XGBoost) by F1-score - Persist and redeploy the updated model and encoders

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	A single, clearly-labelled form (grouped into Personal, Financial, Family and Credit History sections) so applicants can complete it without confusion.
NFR-2	Security	Form inputs are validated server-side; no sensitive applicant data is stored beyond the request lifecycle; HTTPS is enforced via the Render hosting platform.

NFR-3	Reliability	The model and encoders are version-locked pickle artifacts, so predictions are consistent and reproducible across requests and deployments.
NFR-4	Performance	Prediction inference completes in well under a second per request, since the model is loaded in memory and no external network calls are made during scoring.
NFR-5	Availability	Hosted on Render with automatic restarts on crash; designed to be reachable as a live demo with minimal manual intervention.
NFR-6	Scalability	The stateless Flask design allows additional instances to be added behind Render's infrastructure as applicant volume grows.